

Business Problem Statement

A mid-sized pizza restaurant chain is seeking to improve its sales performance and operational decision-making. Management has observed fluctuations in daily sales, differences in product popularity, and varying customer purchasing behavior across pizza categories and sizes. However, the company lacks a consolidated analytical view to clearly understand which products drive revenue, when peak sales occur, and how customer demand varies over time.

The management team wants to leverage historical pizza sales data to gain actionable insights that can support pricing strategies, menu optimization, and sales planning. They are particularly interested in identifying top-selling pizzas, revenue contribution by category and pizza, and time-based sales trends.

You are tasked with analyzing the pizza sales dataset and answering the following overarching business question:

“How can the company use pizza sales data to identify sales trends, optimize product performance, and support data-driven business decisions?”

Deliverables

1. Data Preparation & Database Setup (SQL Server – SSMS)

- Import raw pizza sales data from Excel into SQL Server
- Create a structured database and tables
- Ensure data consistency and readiness for analysis

2. Data Analysis (SQL)

- Write SQL queries to analyze:
 - Total revenue and total orders
 - Sales performance by pizza category and size
 - Best-selling pizzas by quantity and revenue
 - Sales trends over time (daily/monthly patterns)
- Generate aggregated datasets to support reporting and visualization

3. Visualization & Dashboard (Two Versions)

- **Version 1: SSMS + Excel**

Connect Excel to SQL Server to clean, analyze, and visualize pizza sales data. Built an Excel dashboard to present key sales metrics and trends.

- **Version 2: SSMS + Power BI**

Connected Power BI to SQL Server to process data and build an interactive dashboard highlighting sales performance and time-based trends.

Both version deliver similar insights and KPIs, demonstrating consistent analytical outcomes using different visualization tools.